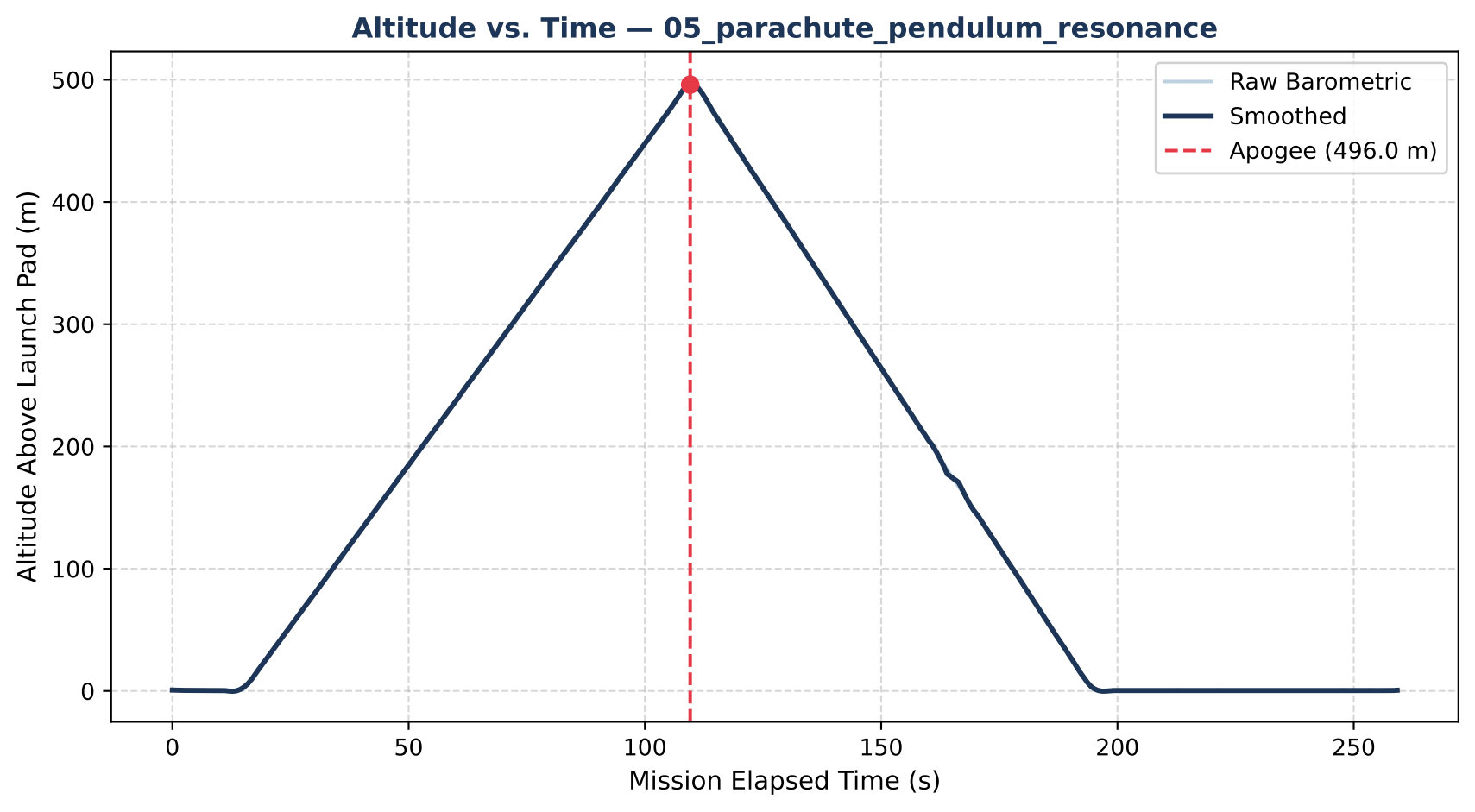
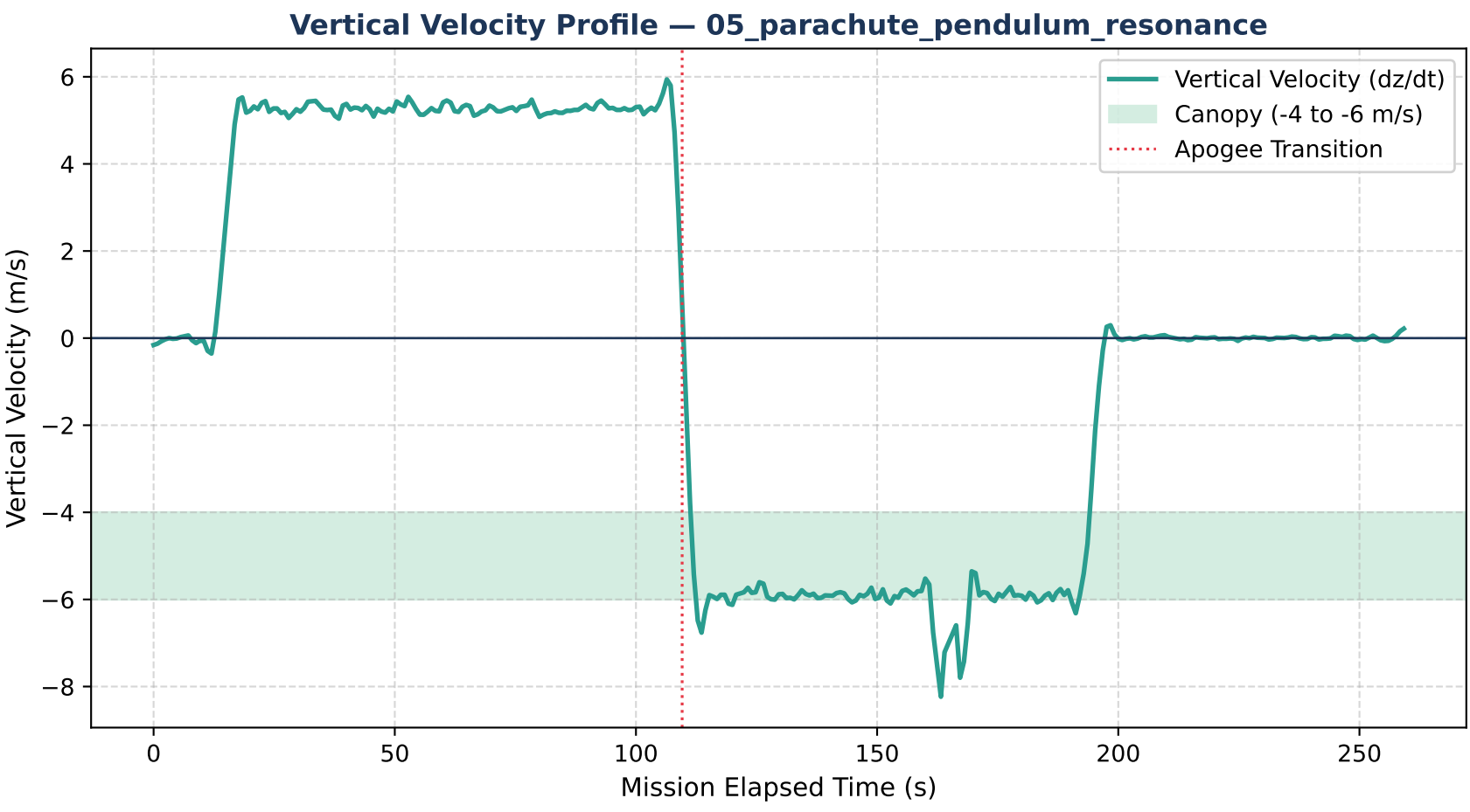
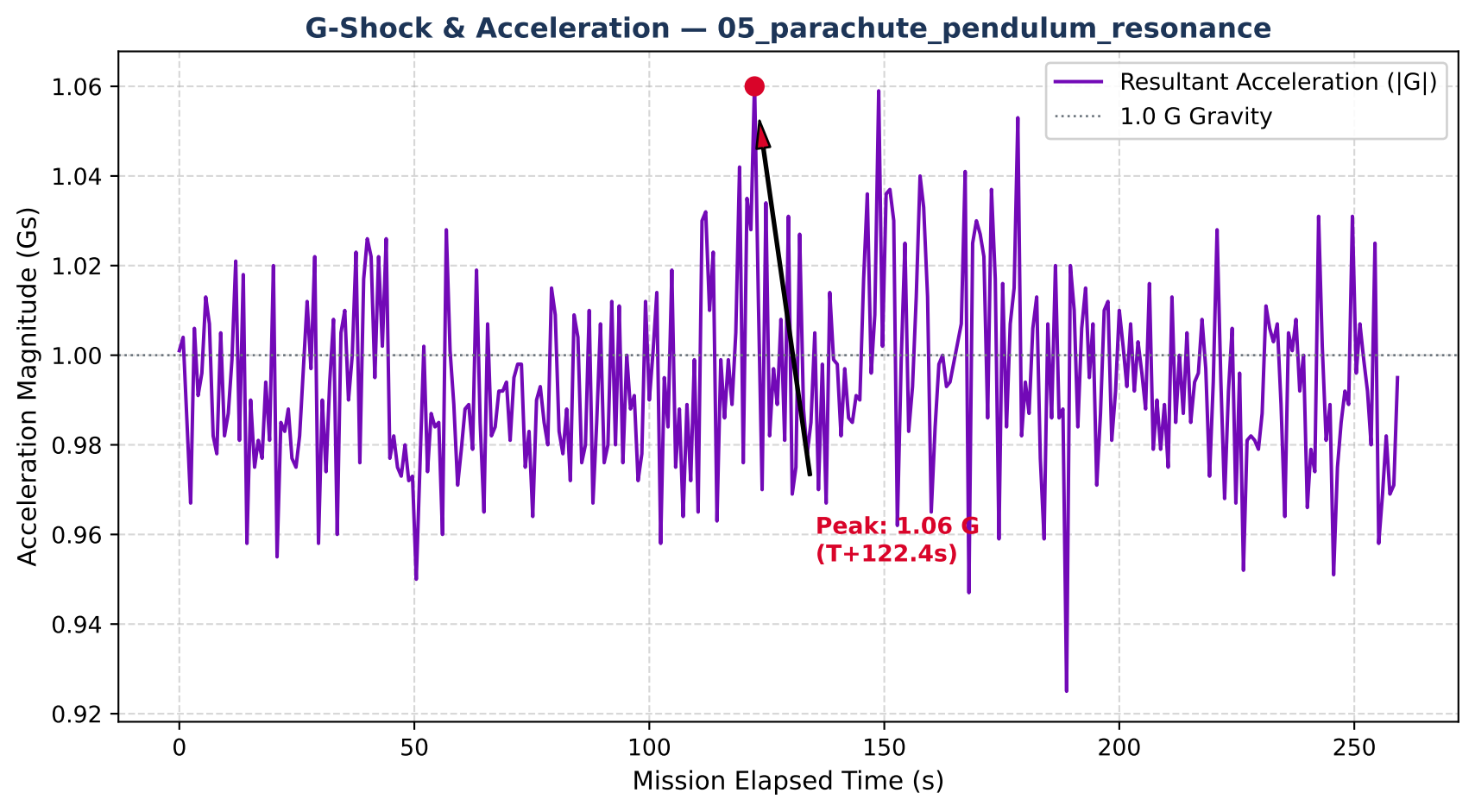
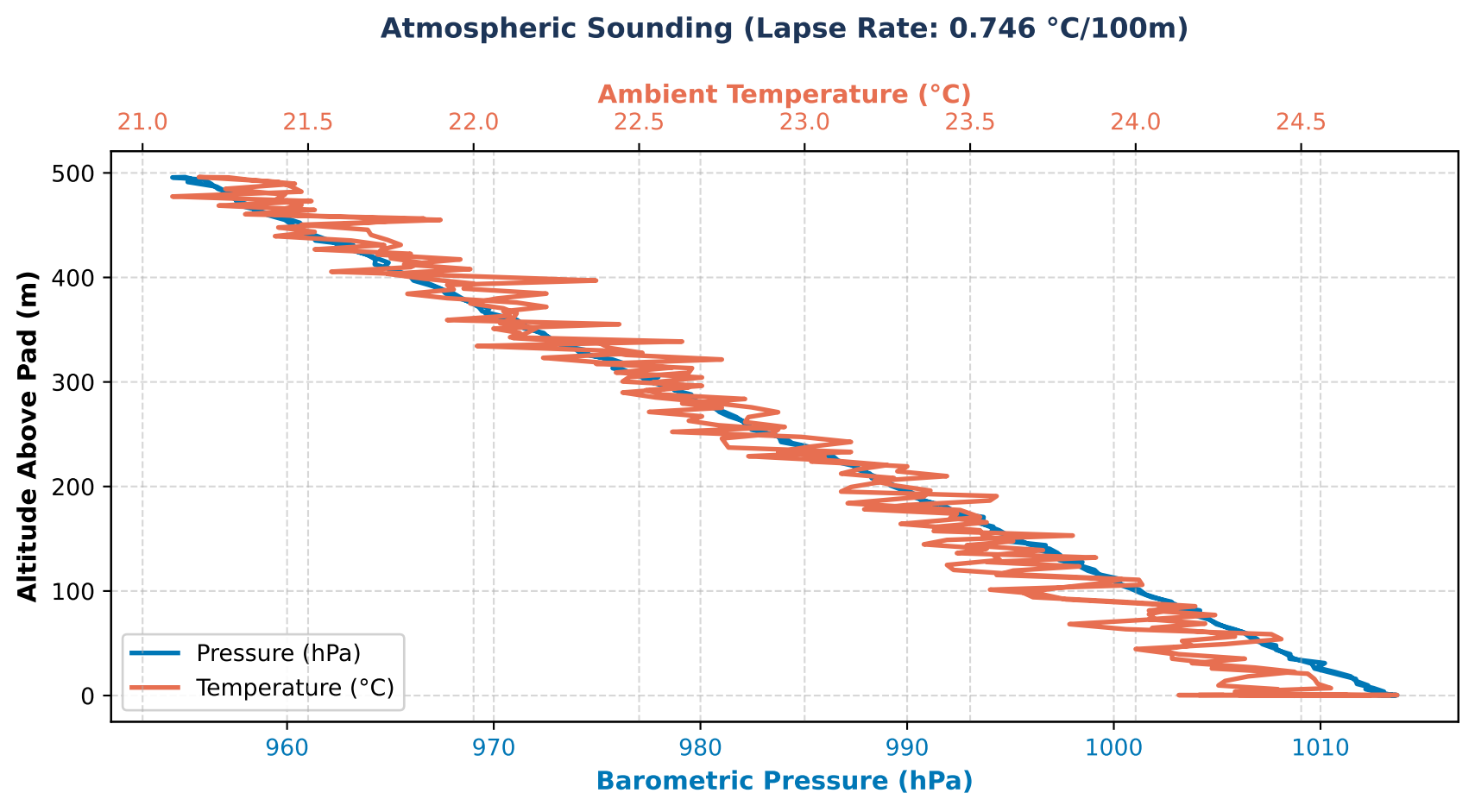
Cognitive CanSat - Mission Summary: 05_parachute_pendulum_resonance

Flight Dynamics Parameter	Calibrated Value
Mission Name	05_parachute_pendulum_resonance
Apogee (m)	495.98
Time to Apogee (s)	109.58
Max Ascent Velocity (m/s)	5.94
Max Descent Velocity (m/s)	-8.23
Max G-Shock (Gs)	1.06
Touchdown Shock (Gs)	1.03
Mission Duration (s)	259.19
Min Temperature (°C)	21.09
Max Temperature (°C)	24.79
Min Pressure (hPa)	954.45
Max Pressure (hPa)	1013.71
Lapse Rate (°C/100m)	0.746
Telemetry Frames	323
Landing Lat	22.574818
Landing Lon	88.367978


Altitude vs. Time — 05_parachute_pendulum_resonance
Raw Barometric
Smoothed
Apogee (496.0 m)
Altitude Above Launch Pad (m)
Mission Elapsed Time (s)
0
100
200
300
400
500
0
50
100
150
200
250


Vertical Velocity Profile — 05_parachute_pendulum_resonance
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
6
4
2
0
−2
−4
−6
−8
0
50
100
150
200
250
Mission Elapsed Time (s)


G-Shock & Acceleration — 05_parachute_pendulum_resonance
Resultant Acceleration (|G|)
1.0 G Gravity
Peak: 1.06 G
(T+122.4s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
1.06
1.04
1.02
1.00
0.98
0.96
0.94
0.92
0
50
100
150
200
250


Atmospheric Sounding (Lapse Rate: 0.746 °C/100m)
Ambient Temperature (°C)
21.0
21.5
22.0
22.5
23.0
23.5
24.0
24.5
Altitude Above Pad (m)
500
400
300
200
100
0
Pressure (hPa)
Temperature (°C)
960
970
980
990
1000
1010
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 05_parachute_pendulum_resonance

